

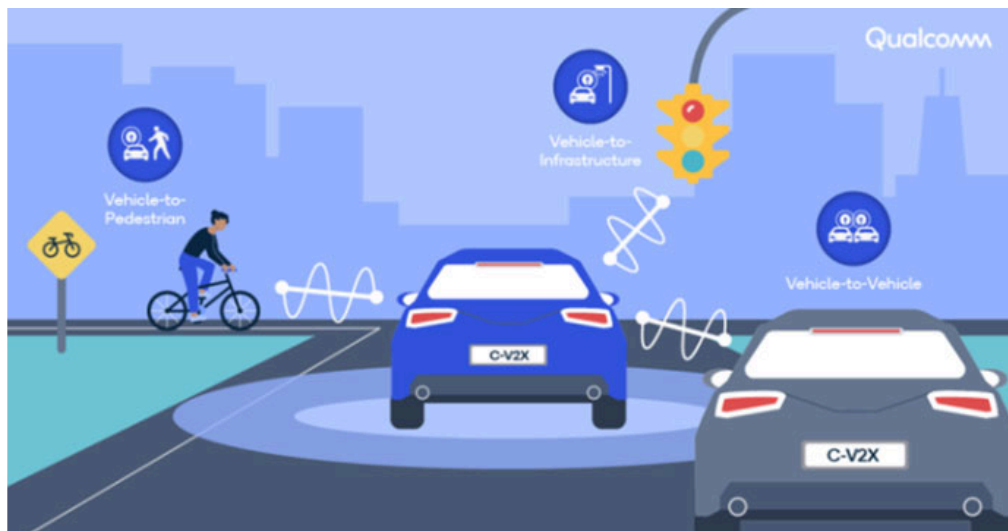
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C-V2X delivers outstanding performance for automotive safety



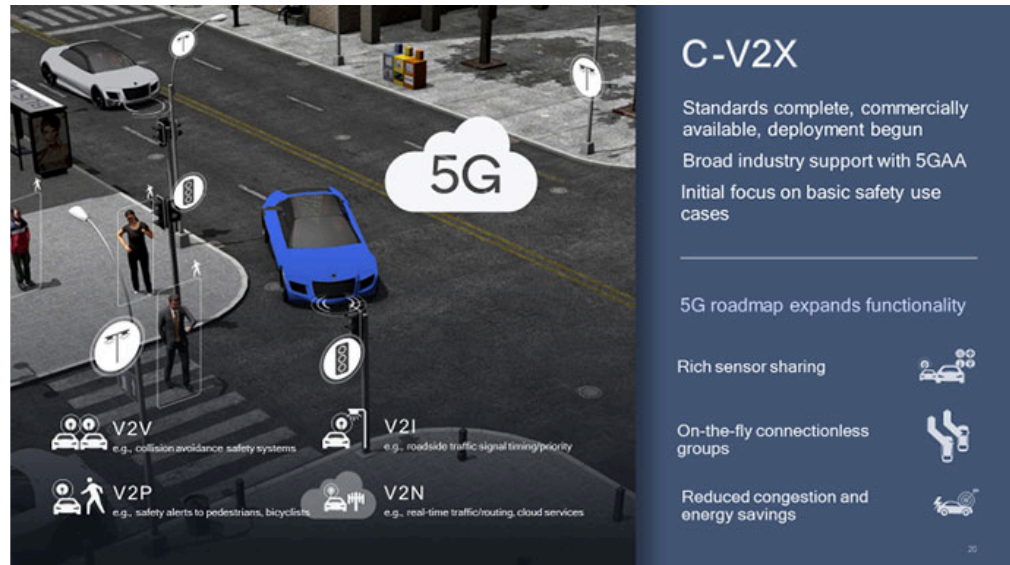
Written by
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As automakers and regulators move closer to complete adoption of connected vehicles, and cities and other road owners ambitiously pursue smart and safe transportation, it is important to use a high

C-V2X, which includes vehicle-to-vehicle (V2V), vehicle-to-pedestrian (V2P), and vehicle-to-infrastructure (V2I), builds on several decades of research and standardization work in wireless technologies, automotive safety, and transportation efficiency. It includes a direct communication mode, called sidelink, which allows vehicles and roadside units (RSUs) to communicate reliably and directly with each other without needing the cellular network.



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Leading automakers worldwide have acknowledged the benefits of C-V2X sidelink. Various New Car Assessment Programs (NCAP)s around the world are considering C-V2X as they look at the safety impacts it can offer. China is rapidly working toward a massive deployment of C-V2X vehicles and infrastructure. Europe too has defined a new European Standard (EN) for the use of C-V2X as an access layer technology for the Intelligent Transportation System (ITS), developed by the European Telecommunication Standardization Institute (ETSI). In the meantime, initial deployments in

20 MHz can be used for mobility enhancing messages and applications. The use of 20 MHz for basic safety messages is in accordance with the latest J3161/1 standard that was leveraged from the earlier V2V applications standards for Dedicated Short-Range Communications (DSRC). It is worth noting that 5GAA¹ studies are in general agreement with the Car 2 Car Communication Consortium, but that the core basic safety messages can be implemented in 20 MHz.

Enhanced road safety is the basic requirement driving most, if not all innovations in the automotive world. Reliable and timely radio performance are uncompromisable for any solution introduced on our roads. Given Qualcomm's 35-year history of innovation and leadership in wireless communication, pioneering efforts in defining 3GPP standards, and two decades of providing the automotive industry with state-of-the-art telematics solutions, [we know that C-V2X, which was first introduced in 3GPP Release 14, is a far superior and modern wireless technology](#). We are driving a rich roadmap of 5G technologies in 3GPP, which we are leveraging for C-V2X to create a strong evolution path that supports future, advanced safety use cases.

Utilizes 5G NR flexible framework

Scalable OFDM-based air interface



Such as wideband carrier support (>20 MHz) and different sub-carrier spacing

Flexible slot-based framework



Such as adding sidelink and dynamic reference signal for various speed

Advanced channel coding



State of the art LDPC/polar coding to deliver performance

Leverages LTE V2X concepts



5G
V2X

Introduces advanced capabilities

- Efficient sidelink link level design for optimized performance at all speeds
- Connectionless 'on-the-fly' distance-based groups
- Multicast with distance-based reliability and application relevancy

And increased performance

- Lower latency
- Higher spectral efficiency
- Higher capacity

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performance tests were conducted at Crash Avoidance Metrics Partners LLP (CAMP) over the last decade to evaluate the two radio technologies.

Under closely monitored conditions, field tests were conducted under a variety of environmental conditions, both ideal and adversarial. The tests were conducted using packet length of 193 bytes for basic safety messages, transmitted on ITS band (channel 184) with bandwidth of 10 MHz. The detailed test report² from 5GAA elaborates on the test set up, conditions, parameters, and results. However, look at the overall C-V2X performance in comparison to DSRC, synthesize key reliability and link margin performance metrics, and share the findings of the near-far problem for C-V2X. C-V2X clearly demonstrated superior performance and reliability in comparison to DSRC. These tests were designed to represent real-world traffic conditions for non-line-of-sight (NLOS) static and moving obstructions and included background noise and interference. C-V2X showed significant range (link margin) advantage over DSRC, which translates to enhanced safety for drivers and vulnerable road users (VRU). And yes, the results did not surprise us since these performance characteristics were part of our requirements when designing C-V2X and were already anticipated from modeling and lab results.

Reliability and link margin

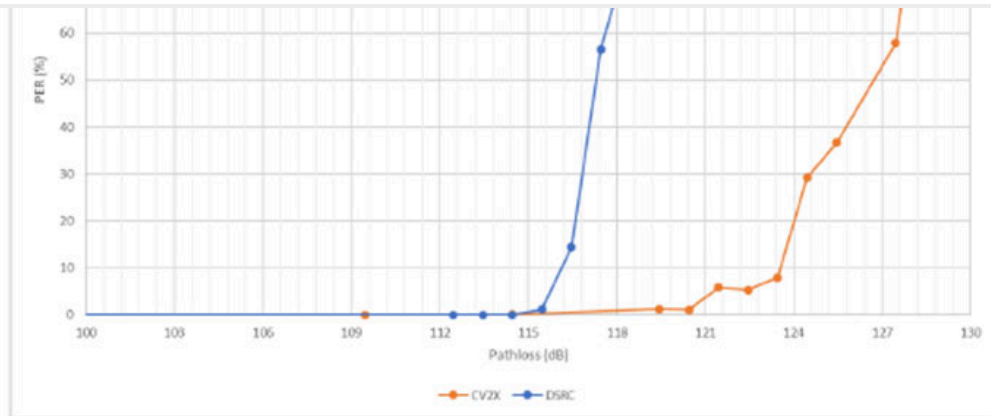
Systematic tests modeling path loss with and without background noise were conducted to assess communication reliability between transmitting and receiving vehicles at various signal strengths. C-V2X

Imagine your car is transmitting a Do Not Pass Warning (DNPW) signal to a receiving vehicle on a hilly road.

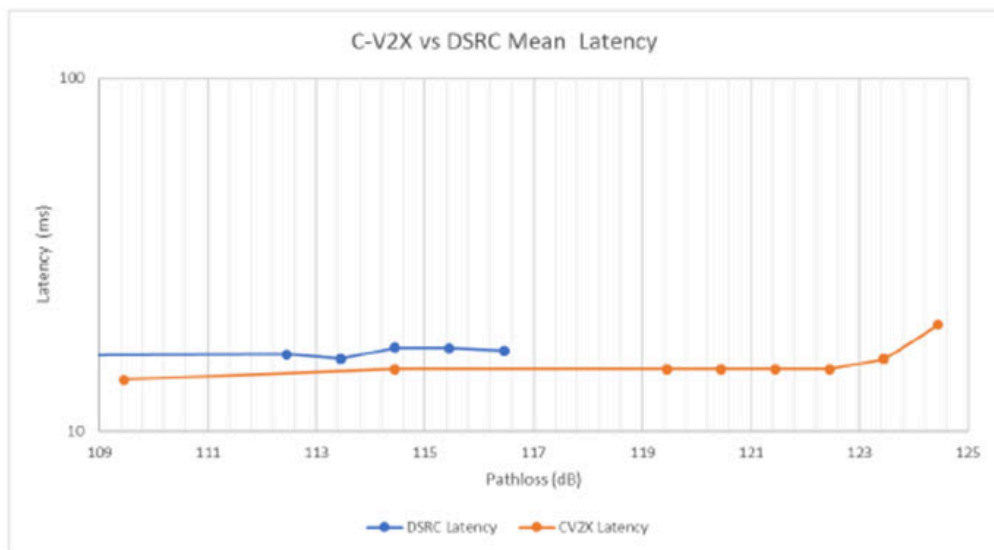
Low signal strength can affect the receiving vehicle's ability to get the signal on time, potentially leading to a collision. However, with superior link performance, C-V2X can alleviate the condition of poor signal strength, which DSRC cannot.

To deliver high reliability, C-V2X sidelink uses blind hybrid automatic repeat request (HARQ), where the same data packet is transmitted with a different coding. The receiving vehicle can reconstruct the original message using both (or multiple, as the case may be) of the transmitted messages. *DSRC uses no form of HARQ and the result is that C-V2X sidelink has an advantage of ~ 10 dBm in link budget over DSRC.* It should be noted that C-V2X can transmit over a longer time but uses narrower bandwidth without loss in system capacity by leveraging sub-channel sharing with other transmissions. This provides C-V2X with ~2x link budget gain over DSRC.

Here are two insightful graphs for reference from the test report² that show comparisons between DSRC and C-V2X. The graph below demonstrates C-V2X having significantly lower packet error rates (PER) as interference increases.



The second graph below demonstrates another key characteristic — mean latency, in increasing interference levels resulting in path loss. C-V2X shows consistently lower mean latency in comparison to DSRC even as path loss increases.



Thus, in the presence of path loss and channel noise, the performance advantage of C-V2X over DSRC becomes even more significant. Real-world conditions more often than not consist of NLOS conditions. As current on-board sensors such as LiDAR and camera

Unlike DSRC, in C-V2X, users can share the channel at the same time, in other words, transmit simultaneously on adjacent sub-channels. But some have raised concerns that this may cause a near-far problem, meaning, that it would impact a receiving C-V2X vehicle's ability to detect a weaker incoming signal due to obstructions such as tall buildings, trees, and other vehicles or interference from a stronger signal in an adjacent sub-channel. So, we conducted a test to assess the performance of a C-V2X vehicle receiving basic safety message signals from two vehicles transmitting at different power signal levels in adjacent subchannels over a 10 MHz channel in the ITS band. The first vehicle transmitted almost every 1 ms using the first half of the frequency resources and the second vehicle transmitted every 100 ms using the second half of the frequency resources. The receiving vehicle is also configured to receive on an ITS band of 10 MHz. Test results showed that for all practical road scenarios, C-V2X meets the minimum requirements set by the 3GPP Release 14 standards keeping the average leakage of the device under test ~ -35 dB.

[Related congestion tests were conducted which confirmed that the channel utilization provided an efficient use of the channel](#) under test while maintaining a low packet error rate to improve overall performance in a congested environment. Even in a poor communication environment, high-priority basic safety messages for congested and collision scenarios showed higher reliability in performance thanks to how the C-V2X resource selection algorithm works to protect high-priority safety messages.

commonplace, especially as countries expand their economies and continue to grow³. Any technology that is deployed on our roads must be equipped to handle such congestion levels. Imagine driving around downtown at an intersection with a popular corner café offering free Wi-Fi. With the Wi-Fi interference and obstructions from tall buildings or foliage, you would still want your car to receive and transmit reliably and communicate early alerts to avoid collisions. We tested C-V2X for these highly congested settings, and end-to-end latency still remained within 100 ms. Additionally, we tested for varying communication environment conditions and overall C-V2X has 1.3 to 2.9 times the range advantage over DSRC. The table below shows a range comparison of both the technologies to show you in perspective the clear advantage that C-V2X has over DSRC. What should really catch the eye is the significant link budget gain for C-V2X in NLOS scenarios. To reiterate, NLOS scenarios are a close replica of real-world traffic and C-V2X offers a benefit of more than twice the range which can rapidly translate to massive safety improvements as we work toward a more connected and intelligent transportation network on our roads. The table below is again derived from the test report and it provides a quick range comparison of DSRC and C-V2X at 11 dBm transmit power.

NLOS blocker (5GAA)	625/725	425
NLOS intersection	600/800	90/400
Co-existence with Wi-Fi	950	550

So, what's next?

[Our 5G vision](#) is for the world to be connected with a common, unifying connectivity fabric supporting various devices across diverse deployments and spectrums. Sidelink continues to be a key focus area of our innovation. While C-V2X sidelink in 3GPP Release 14 provided the reliability and performance for basic safety messages, the C-V2X sidelink extension in 3GPP [Release 16+ builds on the strong foundation of our core 5G technologies](#) to bring higher throughput, lower latency, and reliable multicast communication, for advanced safety applications. It is noteworthy that vehicles equipped with Release 16-based 5G V2X technology will also support C-V2X sidelink from Release 14/15 for basic safety applications.



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Release 16 introduced reliable multicast communication for the C-V2X sidelink, which is enabled by key foundational innovations such as utilizing distance as a new paradigm in the physical layer and addressing application-specific communication using NACKs. Using distance as a dimension, helps to achieve uniform communication range for both LOS and NLOS scenarios. Groups can form on-the-fly for exchanging messages with little to no overhead for group formation and dismantling. The superior reliability, extended range, low latency, and NLOS capabilities of 5G V2X allows it to support advanced safety use cases at higher vehicle speeds (~500kmph) and challenging road and traffic conditions. 5G Release 16 sidelink includes advanced positioning technology which comes specifically handy in poor GNSS reception areas such as a tunnel or a parking garage. Vehicles can continue to reliably communicate using 5G V2X sidelink for safe driving.

Our innovations in 5G V2X sidelink are setting the stage well for an enhanced autonomous driving experience, faster and efficient traveling, and reduced carbon emissions in addition to enhanced road safety for everyone. We are designing a two-stage control



Slot structure showing two-stage control



- Stage 1 format for resource allocation
- Stage 2 format for a Rel 16 application
- Stage 2 format for a Rel 17 application

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The significance of sidelink continues even in future 3GPP releases as we strive to [extend sidelink capabilities from automotive to public safety and beyond](#).

Conclusion

In light of the FCC's proposed transition phase from DSRC to C-V2X, the technologies can be separated by geography. C-V2X can be exercised in the U.S. with experimental license right now, giving road infrastructure owner-operators a head start on broad deployment of this critical safety technology. It is therefore important to understand [in addition to the products available](#), that we cut through the haze of

allow the U.S. to emerge as a market leader in the transportation ecosystem. As 5G research and penetration continues at breakneck speed across industries, C-V2X has a clear path for evolution to 5G V2X. This is important for automakers, infrastructure owner-operators, and the rest of the ecosystem wanting to maintain an edge over the competition in an extremely challenging environment. And while we are focusing on reliable connectivity on our roads, let's be mindful that enhanced safety is the top priority.

1. 5GAA is a multi-industry global organization with more than 140 companies, including some of the most recognized leaders in automotive and telecommunications industries.

2. https://5gaa.org/wp-content/uploads/2018/11/5GAA_P-190033_V2X-Functional-and-Performance-Test-Report_final-1.pdf

3. https://www.tomtom.com/en_gb/traffic-index/ranking/

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